Physica A 655 (2024) 130195

Contents lists available at ScienceDirect

Physica A: Statistical Mechanics and its Applications

journal homepage: www.elsevier.com/locate/physa

An unsupervised group detection method for understanding group dynamics in crowds

Nipun Choubey [a], Ashish Verma [b,*], Anirban Chakraborty [c]

[a] Robert Bosch Centre for Cyber-Physical Systems, IISc, Bangalore, India
[b] Department of Civil Engineering, IISc, Bangalore, India
[c] Department of Computational and Data Sciences, IISc, Bangalore, India

ARTICLE INFO

Keywords:
Group detection
Group dynamics
Crowd dynamics
Mass religious gatherings
Pedestrian movement

ABSTRACT

Pedestrian groups arrive in large numbers in crowd gatherings, especially of a spiritual nature. Various studies have been done on crowd control in public spaces by analysing the behaviour of pedestrian groups. Understanding group dynamics can help better plan pedestrian facilities and large events. Many existing group sensing models primarily determine social bonding between pedestrians using spatiotemporal parameters, such as distance, directional movement, and overlapping time. However, social bonding determined based on these parameters assumes the bonding to be symmetric, spatially and temporally static and is unaffected by neighbourhood. Our study addresses the issue by relaxing such assumptions and developing an unsupervised group detection model based on potential candidates. The proposed model can handle temporal and spatial variations more effectively than those based on simple spatiotemporal parameters. The model developed is assessed both quantitatively and qualitatively. New metrics are introduced for quantitative evaluation, comparing predicted groups and ground truth instead of pedestrian pairs with ground truth. A visualisation method is developed for the qualitative assessment. Group splits and group merges are calculated to assist in understanding crowd movement patterns. Overall, this study helps in further exploring and assessing groups, which can improve understanding of crowd dynamics.

1. Introduction

Public spaces like railway stations, bus stops, airports, religious sites, malls, stadiums, parks, etc., often encounter large crowds. One of the critical roles of facility managers in these locations is to ensure a smooth flow of pedestrians and avoid any haphazard that can lead to casualties or injuries. An essential aspect of crowd management is understanding pedestrian group behaviour. Pedestrian groups arrive in high proportion in crowd gatherings [2,17]. These groups may comprise family, friends, and acquaintances. Most of the times social groups are found to move cohesively as a single social unit. However, these groups may exhibit behaviours like convergent, divergent, static, or dynamic and may form circular, V, inverted V, linear, and/or river-like patterns. The number of group members negatively correlates with walking speed [13,15,19] and spatial cohesion [6]. It was observed that the evacuation time significantly increased with the number of pedestrian groups in the crowd [10]. Apart from being present in large proportions, groups behave differently from individuals. It was observed that the presence of serpentine groups in the crowd increases the crowd risks [18].

* Corresponding author.
E-mail address: ashishv@iisc.ac.in (A. Verma).

https://doi.org/10.1016/j.physa.2024.130195
Received 14 June 2024; Received in revised form 28 August 2024;
Available online 30 October 2024

Thus, understanding the group dynamics could assist in better planning of pedestrian facilities and large events.

Most of the studies on pedestrian group sensing consist of two key steps. The first step involves defining social bonding among all the pedestrians. The social bonding is calculated for each pair of pedestrians present in the scene. Here a pedestrian pair is referred to any combination of two pedestrians. Clearly, if there are P pedestrians at a given scene at time t, then there are $\binom{P}{2}$ pedestrian pairs at that time. Some studies filter out pedestrian pair based on the social bonding while others filter pedestrian pairs by defining a function that estimates the probability of social bonding between a pair of pedestrians. This function is referred as social bonding probability function (SBP) in this paper. The second step involves combining these pedestrians and obtain groups based on the SBP function. A pedestrian pair is generally classified as a group pair (both pedestrians belong to same group) or a non-group pair (both pedestrians are not from same group). Various studies used spatiotemporal parameters such as distance, time, and angle to estimate the SBP function between pedestrians. Spatiotemporal-based SBP functions are symmetric in nature and hence, are maladaptive to local surroundings and are often resistive to temporal variation. Local surroundings around a pedestrian pair play a key role in determining whether the pedestrian belongs to a group. To make readers understand, an example is illustrated in Fig. 1. In Fig. 1, circles represent pedestrians, their number represents pedestrian ID, and the colour represents their group. Pedestrians 1, 2 and 6, 7, 8 are two groups, while pedestrians 3,4,5 are individuals. The distance between pedestrians 1–2, 3–4 and 5–6 are the same. Although pedestrians 1–2 are in the group while others are not. In this case, SBP function based on distance would have same value and thus model would categorise all three pedestrian pairs into the same category. Similarly, SBP function based on other spatiotemporal parameters are also symmetric in nature and invariant to the local situation, resulting in a similar classification issue.

This study proposes an unsupervised group detection model with a novel rankSBP function that assigns the social bonding probability to each pedestrian based on their relative order. rankSBP is adaptive to local surroundings and temporal variation and considers best N candidates for each pedestrian to form groups. Unlike existing literature that evaluates the model by comparing class of pedestrian pairs (group or non-group) alone, this study introduces new metrics that evaluate the models' performance based on the predicted groups. Apart from quantitative evaluation, the proposed group detection model is assessed qualitatively. A visualisation method is developed for qualitative assessment, combining model predictions with ground truth data. Further, this study estimates that group splits and group merges to understand crowd movement better.

The rest of the paper is structured as follows. Section 2 summarises related works done in group detection. Section 3 explains the method adopted in detail and Section 4 describes the data used in the study and the evaluation metrics introduced. Section 5 assesses the model performances quantitively and qualitatively and discusses the inferences, followed by the conclusion section, that summarises the study's findings.

2. Literature review

Several parameters are explored by researchers for defining the social bonding of a pedestrian pair. Studies used instantaneous spatiotemporal parameters such as pairwise distance [1,5,12,14,16], temporal overlap [1,12,14], direction of movement [12,16,21], relative velocity [11,5] which are averaged over time. Other studies used parameters such as Hausdorff distance [1,22] between pedestrian trajectory, gazing direction [8], visual descriptors, trajectory similarity [9], velocity similarity [22] based on time series, common starting & ending location [20] of pedestrian pair.

Earlier studies evaluated their group detection models on SBP value calculated for each pedestrian pair using metrics like accuracy [12,21], false positives/precision [12,22], recall [22], and F1 Score [22]. While the final output of any group detection model is supposed to be a group. A few studies evaluated their model on the final output, i.e., groups. Ge et al., [5] evaluated their models based

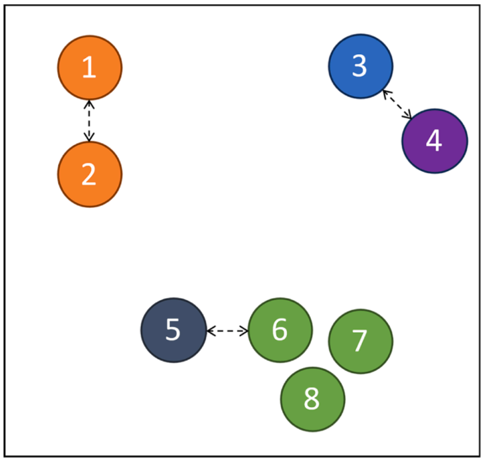


Fig. 1. An Illustration of pedestrians (circles) present in an environment, the number represents the ID, and the colour represents the group they belong to.

on Dichotomous and Trichotomous metrics. The dichotomous metric classifies a pedestrian as a group member or an isolated pedestrian. Meanwhile, the trichotomous checks the size of the group to which a pedestrian belongs. Mazzon et al., [11] coined a new metric called Group detection success rate, which considers the predicted group as correct if 60 % of the members matches with the ground truth. Cheng et al. [1] used the group's IoU (Intersection over Union) to evaluate their model. They reported the mean and the standard deviation of IoU over pedestrians.

Many existing studies developed their models based on spatiotemporal parameters. An advantage of these models is that they are simple to create and interpret. However, determining the social bonding based on spatiotemporal parameters assumes that a pair of pedestrians' bonding is symmetric, static, and not influenced by surroundings. This assumption is relaxed in this study, and an unsupervised group detection model based on potential candidates is developed. The model uses rankSBP function which is asymmetric, considers the surroundings and updates over time.

Most of the studies assess their models based on pedestrian pair classification, an intermediate result of the group detection model. However, the final output of the group detection model is the groups predicted by the model. While some studies have assessed their model based on group predictions, their assessment is based on the number of groups with a high overlap with ground truth. We consider this assessment as incomplete and needs more important details. Thus, this study introduces three new evaluation metrics for evaluating the performance based on the final output of group detection models, i.e., groups. This study also analyses the joining and leaving of pedestrians to/from the groups to understand group movement patterns.

3. Methodology

This study proposes an unsupervised group detection model that considers the relative ordering of pedestrians to form groups. The methodology flowchart is illustrated in Fig. 2. At timestep t, there are n pedestrians. The set of pedestrians is denoted by $P_t = \{a_1, a_2, \ldots a_n\}$, where a_1, a_2 are pedestrians with ID 1, 2 etc. The algorithm first identifies potential candidates for each pedestrian and estimates the probability, followed by formation of groups. Finally, groups association is carried out to map groups predicted at timestep t with groups predicted in timestep $t-1$. Same procedure is repeated at each timestep. Each step is described in the following subsections.

3.1. Identifying potential candidates

The $rankSBP_i$ function estimates the probability based on the social bonding rank of all the pedestrians with respect to pedestrian i. The social bonding $(f_i(j,t))$ of pedestrian j with respect to pedestrian i is defined as

$$f_i(j, \quad t) = \frac{\overline{d}_{ij}}{T_{ij}} \tag{1}$$

where, $\overline{d}_{ij}$ is the mean distance between pedestrians i and j until time t and T_{ij} is the common time for which both pedestrians were present together until t.

This study considered mean distance and total common time to ensure spatial proximity and temporal continuity. All pedestrians are ranked based on the estimated social bonding with respect to pedestrian i. The $rankSBP_i$ value for pedestrian j having rank $r_{j,i}$ with

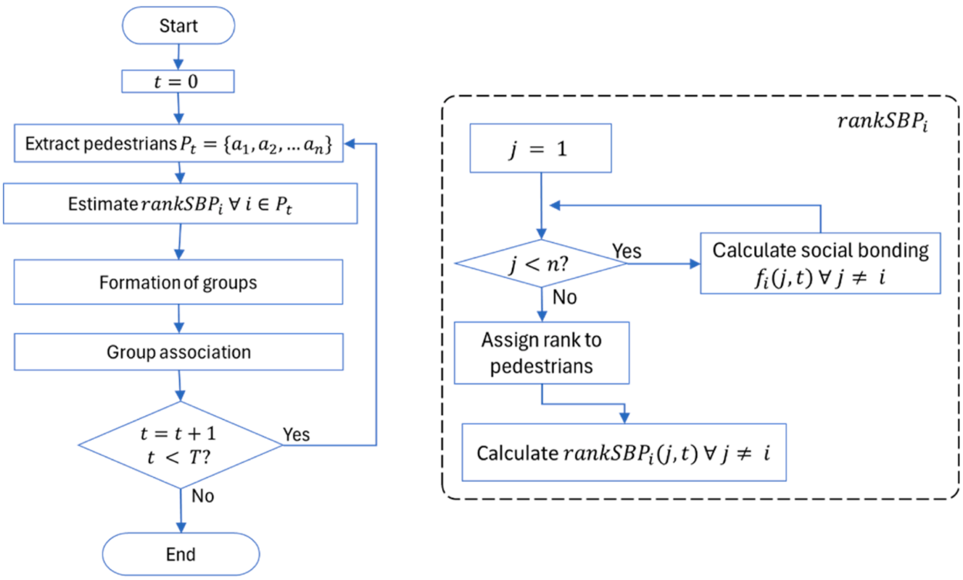


Fig. 2. Unsupervised group detection model workflow.

respect to pedestrian i is defined in Eq. (2).

$$rank\mathrm{SBP}_i(j;N) = \begin{cases} \dfrac{2 * r_{j,i}}{N(N+1)} & for\ 1 \le r_{j,i} \le N \\ 0 & otherwise \end{cases} \tag{2}$$

$$\sum_j rank\mathrm{SBP}_i(j;N) = 1 \tag{3}$$

In Eq. (2), N is the maximum number of potential candidates considered. The $rank\mathrm{SBP}_i(j;N)$ gives the probability of pedestrian j having social bonding with pedestrian i considering maximum N candidates and hence, the sum of probabilities of all pedestrians with respect pedestrian i is 1 (Eq. (3)).

3.2. Group Formation

Based on the rankSBP values obtained in the previous step, pedestrians are clustered into groups with agglomerative clustering. Since agglomerative clustering requires a symmetric metric, the average of rankings is used for clustering. Clusters formed with multiple pedestrians are treated as groups, while clusters containing single pedestrian are treated as isolated pedestrians or individuals.

3.3. Group Association

Group association is a critical step in group tracking. In this step, the groups are detected in timestep $t-1$ are mapped to groups in timestep t. For mapping groups across timesteps, we consider the pedestrians common between the groups at the timestep $t-1$ and t. Let the groups at $t-1$ timestep be $G_{1,t-1}$, $G_{2,t-1} \ldots G_{i,t-1}$ and groups at timestep t be $G_{1,t}$, $G_{2,t} \ldots G_{j,t}$. A group $G_{i,t-1}$ is mapped with a group G_t if all the pedestrians are common in both groups. With this association, the group trajectories can be obtained. However, there may be few groups at timestep t which may not completely overlap with any group in timestep $t-1$. This study considers these unmapped groups as new groups. Once group association is done, group trajectories can be extracted. Group count and group speed can be estimated from the group trajectories.

3.4. Identifying splitting and merging of groups

Once group association is done, group trajectories can be extracted. Although all the groups may not be mapped, this results in new group formation. These new groups may formed due to new pedestrians entering the scene or existing pedestrians with different compositions. In the latter case, pedestrians in one group at timestep $t-1$ are split into multiple groups at timestep t or multiple groups in timestep t are merged into a single group.

Two terms, namely, group merge and group split, are introduced to capture the movement of pedestrians among groups. The study defines a group merge as a situation when pedestrians from two or more groups at timestep $t-1$ are clustered into single groups at timestep t. Similarly, when pedestrians from one group at timestep $t-1$ is clustered into two or more groups at timestep t then it is considered a group split. Group split and group merge can be due to various reasons, such as geometry constraints, time constraints, overtaking other pedestrians, allowing other pedestrians to pass in between, more considerable speed difference between group members, etc.

Since the proposed model considers pedestrians in spatial proximity and temporal continuity to form groups, a higher number of group merges indicates a converging crowd, and a higher number of group splits indicates a diverging crowd. However, higher values for group merge and group split suggest that the crowd movement is less organised.

4. Evaluation metrics

Unlike previous studies that evaluate the model based on pedestrian-pair classification, this study evaluates the group detection models based on the groups obtained from the model and not at the pair level. New evaluation metrics are defined for comparing groups obtained from the ground truth with groups obtained from the group formation step. The study refers groups obtained from the ground truth as real groups while groups obtained after group formation as predicted groups. To evaluate the performance of the group detection model, the groups predicted by the model needs to be map with groups mentioned in the ground truth. For this, first, we need to define group similarity.

4.1. Group similarity

This study defines group similarity $o_A(B)$ as the similarity of group B with respect to group A. It is the ratio of the number of pedestrians common in groups A and B to the number of pedestrians in the group A.

$$o_A(B) = \frac{|A \cap B|}{|A|} \tag{4}$$

where, $|A|, |A \cap B|$ represents the number of pedestrians in group A and the number of pedestrians common in groups A and B respectively. The value of group similarity would range between 0 and 1 with zero representing no pedestrian is common and one represents all the pedestrians are same in both groups. The group similarity shows the extent of overlap among groups. The expression $o_A(B) = 0.6$ can be read as group B is 0.6 similar to group A or 60 % of pedestrians in group A are found in group B. It is important to note that the group similarity is an asymmetric metric. i.e., $o_A(B) = 0.6$ does not imply $o_B(A)$ is also 0.6. For example, assume group A consists of pedestrians {1,2,3,4,5} and group B consists of pedestrians {1,2,3,6}. In this case, $|A| = 5$, $|B| = 4$, $|A \cap B| = 3$. Also, $o_A(B) = 0.6$ and $o_B(A) = 0.75$.

Based on group similarity, the next task is to find an optimal mapping from set of real groups $\mathbb{R}$ and set of predicted groups $\mathbb{P}$. First, assume given two sets $\mathbb{A} = \{A_1, A_2, \ldots\}$ and $\mathbb{B} = \{B_1, B_2, \ldots\}$, where A_1, A_2 are the groups in set $\mathbb{A}$ and B_1, B_2 are the groups in set $\mathbb{B}$. The optimal mapping of set $\mathbb{B}$ relative to set $\mathbb{A}$ would give the list of pairs of groups $[(A_i, B_j)\ldots]$ for which the sum of group similarity of groups of $\mathbb{B}$ relative to groups of $\mathbb{A}$ is maximum. The optimal mapping $O_{\mathbb{A}}(\mathbb{B})$ is defined by Eq. (5).

$$\mathrm{O}_{\mathbb{A}}(\mathbb{B}) = \mathrm{argmax} \sum\nolimits_{A_i \epsilon \mathbb{A}, \quad B_j \epsilon \mathbb{B}} o_{A_i}(B_j) \tag{5}$$

Identifying optimal mapping is a classic assignment problem [7], where two sets of vertices in a bipartite graph are mapped to get minimum cost. In our case, the two sets of a bipartite graph would be the set of real groups and the set of predicted groups, and the weight of the edge between the vertices would be calculated based on group similarity. We used the linear sum assignment (LSA) [4] algorithm to solve the assignment problem. LSA optimises a cost matrix with rows and columns representing the two sets of vertices, where each cell represents the edge cost between the two vertices.

A list of pairs of groups $[(A_i, B_j)\ldots]$ is obtained from Eq. (5). This list would contain the best pairs of groups, providing maximum overlap. However, it would also include pairs of groups that may have only one common pedestrian or a significantly small number of common pedestrians. Such smaller mappings are not considered as a match, and an overlap threshold (Ω) is defined to filter out such pairs of groups. Mazzon et al., [11] considered a prediction to be correct if the group predicted by the model has a 60 % overlap with the real group. We used the same value as the threshold to define our metrics. Given a set of predicted groups $\mathbb{P}$, a set of real groups $\mathbb{R}$, This study defines three metrics *GroupRecall*, *GroupPrecision*, and *GroupMatchScore*.

4.2. *GroupRecall (G_r)*

GroupRecall is the ratio between the number of pairs having an overlap greater than Ω to the number of real groups. The best $-\mathrm{map}(\mathbb{R}, \mathbb{P})$ provides the set of predicted groups with the highest similarity with the real groups. This metric is sensitive to pedestrians, which are missing in the predicted group but present in the real group. This ratio varies between 0 and 1 with a higher value indicates most of the groups predicted by the model have a higher overlap with the real groups. This ratio accounts for false negatives at the group level.

$$G_r \quad (\mathbb{R}, \mathbb{P}) = \frac{|\{(R_i, \quad P_j) : \quad \forall \quad (R_i, \quad P_j)\epsilon \mathrm{O}_{\mathbb{R}}(\mathbb{P}), \quad o_{R_i}(P_j) > \Omega\}|}{\mathbb{R}} \tag{6}$$

4.3. *GroupPrecision (G_p)*

GroupPrecision is the ratio between the number of pairs having an overlap greater than Ω to the number of predicted groups. The best $-\mathrm{map}(\mathbb{P}, \mathbb{R})$ provides the set of real groups with the highest similarity to predicted groups. This metric is sensitive to pedestrians in the predicted group that are not in the real group. This ratio varies between 0 and 1 with a higher value indicates most of the real groups have a higher overlap with groups predicted by the model. This ratio accounts for false positives at the group level.

$$G_p(\mathbb{R}, \mathbb{P}) = \frac{|\{(P_j, R_i) : \quad \forall \quad (P_j, \quad R_i)\epsilon \mathrm{O}_{\mathbb{P}}(\mathbb{R}), \quad o_{P_j}(R_i) > \Omega\}|}{\mathbb{P}} \tag{7}$$

4.4. *GroupMatchScore (G_m)*

GroupMatchScore provides the number of groups predicted by the model that exactly match real groups. A higher value of GroupMatchScore indicates better performance of the model.

$$G_m(\mathbb{R}, \mathbb{P}) = |\{(R_i, \quad P_j) : \quad \forall \quad (R_i, \quad P_j)\epsilon \mathrm{O}_{\mathbb{P}}(\mathbb{R}), \quad o_{R_i}(P_j) = 1\}| \tag{8}$$

5. Data

This study uses the dataset introduced in [3]. The video data was captured during Mahashivratri 2017 in Ujjain, India. Mahashivratri is a religious festival that occurs every year from February to March. The pedestrians in the video data are walking towards the Mahakaal temple in Ujjain. The dataset contains two clips of length 101 s and 147 s, with 266 and 385 pedestrians annotated, respectively. The clips are termed as clip1 and clip2. The dataset is unique and challenging in multiple ways. First, the data is collected during a religious gathering which has medium density crowd, compared to low density crowd dataset which has been used in earlier studies of group sensing. Second, the placement of camera was constrained due to administrative restrictions and hence, the camera is low mounted and has lower viewing angle. Thus, projective transformation was applied to extract real-world trajectories from pixel-based trajectories. Third, majority of pedestrians were present in the scene for 4–16 s, which is a short duration for group sensing. The snapshots of both the clips are shown in Fig. 3.

Each data point in the pedestrian trajectory consists of information about frame number, pedestrian ID, pixel-based location, pedestrian real-world location (real coordinates), gender, luggage, group ID, whether a pedestrian is stationary/moving, etc. The data annotation/trajectory extraction has been done using the IPS crowd data extraction tool [3]. The visual cues that are used to identify social groups in the dataset are verbal communication, hand holding, direction of travel, proximity, walking patterns, hand gestures, visual appearance, similar luggage. The gender proportion and group size distribution of both clips are mentioned in Fig. 4. It can be observed from the Fig. 4 that there are more males compared to females. Also, the group count decreases with the increase in group size. However, few groups of large size are present in both the clips. As a filtering step, incomplete and small trajectories (<1 s) are removed, and group ID and size are updated for consistency. This step led to 249 and 379 pedestrian trajectories in clip1 and clip2, respectively.

6. Results & discussion

The pedestrian pair classification metrics accuracy and F1 score are used for quantitative evaluation. New metrics introduced in this study are also used for evaluation. Further, for qualitative assessment, a visualisation method is introduced. Finally, group count, group speed, group splits and group merges, are estimated.

6.1. Model evaluation

In this study, the models predicted the groups and tracked after every second. The unsupervised group detection model developed in this study is compared with two existing models that are developed on the same dataset using spatiotemporal parameters. These models are Threshold based model and Linear Probability model [3]. The Threshold based model considers pair of pedestrians as group-pair when the distance between them and difference of angle between their direction of movement is less than the threshold. The Linear probability model considers a linear relationship of probability with these parameters.

The proposed models are compared with these existing models based on accuracy and F1 score and it was found that Candidate models performed better than the existing models (Fig. 5). It can also be observed from the Fig. 5, Candidate model with $N = 3$ performed best among the candidate models. Compared to Threshold based model, the proposed candidate-based models have 9–12 % better F1 score in clip1 and 11–17 % better F1 score in clip2. Compared to Linear Probability model, the proposed candidate-based models have 27–30 % better F1 score in clip1 and 17–23 % better F1 score in clip2. It is important to note that the existing models are supervised model, while the proposed models are unsupervised. This clearly indicates that the Candidate based models are far better than the existing spatiotemporal based models, even though both existing models and proposed models use similar kind of parameters for sensing groups. Hence, considering surrounding pedestrians improves the group sensing models.

The proposed models are evaluated on GroupRecall, GroupPrecision, and GroupMatchScore. The results of the evaluation are mentioned in Table 1 in the form of a fraction (non-reduced form) instead of a decimal value, as the former provides more information. In Table 1, the denominators in G_p column of Table 1 represents the sum of the number of groups predicted by the models at each timestep t, while the numerator represents the sum of the number of groups predicted to have higher overlap with real groups than the overlap threshold Ω at each timestep t. Similarly, the denominators in G_r column represents the sum of the number of real groups at each timestep t while the numerator represents the sum of the number of real groups having higher overlap with the predicted groups than the overlap threshold Ω at each timestep t.

It can be observed by comparing G_p in Table 1, that the candidate N = 3 is having more predicted groups (see denominators). The lower number of group prediction for higher N seems to be intuitive. As models with higher N would combine pedestrians to form more number of large groups resulting in lower overall group count. Compared to other models, Candidate ($N = 3$) prediction is closes to ground truth and hence came out to be the best model for the given dataset.

(a) (b)

Fig. 3. Snapshots of (a) clip1 and (b) clip2.

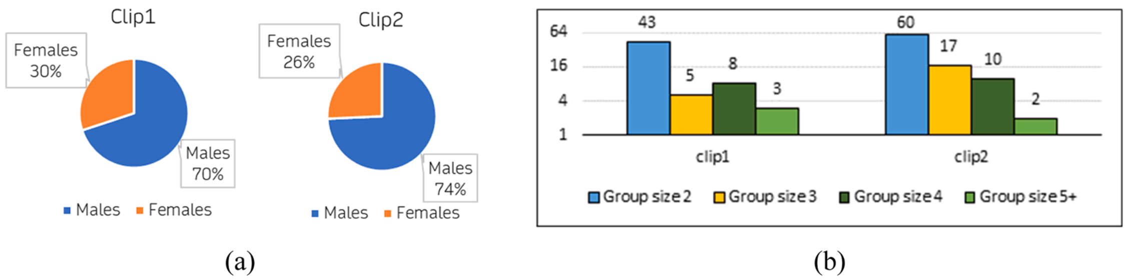


Fig. 4. Clip-wise distribution of (a) Gender and (b) group size.

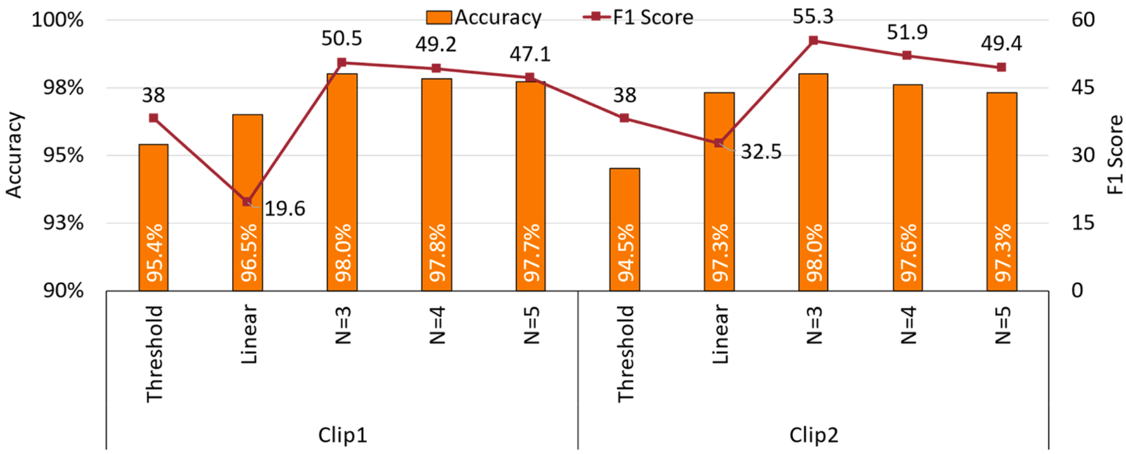


Fig. 5. Performance comparison of Candidate model with existing models based on accuracy and F1 score.

Table 1
Results of modelling after clustering.

Model	Clip1			Clip2		
	G_r	G_p	G_m	G_r	G_p	G_m
Candidate (N = 3)	**1024/1478**	**907/1296**	**653**	**1617/2392**	**1376/2034**	**980**
Candidate (N = 4)	995/1478	864/1247	613	1532/2392	1267/1895	885
Candidate (N = 5)	956/1478	826/1232	574	1452/2392	1169/1779	825

*best values are highlighted in bold.

6.2. Group count and group speed

Table 2 and Table 3 shows the group count and group speed for groups predicted by the proposed model. Table 2 results are consistent with the inferences drawn from Table 1. It can be inferred from Table 2 that the model with more number of candidates combined small groups into large groups. This results in higher count for large groups (group size 5+) and less count for small groups (group size 2, 3) as the number of candidates are increased in the model. The group speeds of the larger group are slightly higher than the smaller group, which opposes the previous findings about groups in shared spaces (Table 3). One of the reasons could be the sense

Table 2
Group count comparison (average).

Model	Group size (Clip1)					Group size (Clip 2)				
	Individuals	2	3	4	5+	Individuals	2	3	4	5+
GT	14	10	2	4	2	14	10	4	3	1
Candidate (N = 3)	10	9	7	4	1	8	11	7	3	1
Candidate (N = 4)	9	9	8	4	1	7	9	8	4	2
Candidate (N = 5)	9	8	8	4	3	7	9	7	4	4

*GT – Ground Truth.

Table 3
Group speed comparison (average).

Model	Group size (Clip1)					Group size (Clip 2)				
	Individuals	2	3	4	5+	Individuals	2	3	4	5+
GT	1.82	1.75	1.85	1.55	1.56	1.86	1.75	1.77	1.55	1.54
Candidate (N = 3)	1.56	1.37	1.67	1.81	1.59	1.7	1.64	1.69	1.68	1.62
Candidate (N = 4)	1.57	1.51	1.81	1.68	1.67	1.68	1.68	1.68	1.63	1.6
Candidate (N = 5)	1.55	1.53	1.84	1.46	1.81	1.7	1.6	1.74	1.57	1.66

*GT – Ground Truth

of urgency among pilgrims to perform their religious activities in the temple. This could be an explanation for observing higher speeds in larger groups.

6.3. Visualization

The proposed group metrics, group count, and group speed provide quantitative insights about groups and the model developed. However, these metrics are not sufficient for capturing minor details. Thus, graph visualisation is developed to assess the proposed models qualitatively.

Fig. 6 provides an understanding of visualisation through a snapshot. The developed visualisation shows the real-world data and the model results in the same animation for each frame. In this visualisation, each frame is equal to the observation area of the study. All pedestrians in that frame are plotted to their respective real-world positions. Each group of pedestrians has a unique combination of shape and colour. Meanwhile, pedestrians not belonging to any group are represented by small circles with a grey colour. The ID of each pedestrian is also displayed for each pedestrian belonging to a group. The potential candidates to each pedestrian are connected through a dotted line, while the solid line connects the pedestrians-based groups predicted by the model.

In Fig. 6, the pedestrians with ID 127,128,129, and 130 are plotted with purple circles, representing that they belong to a single group in the real world. In contrast, the solid line between them represents the group membership link between pedestrians predicted by the model. It clearly shows that the model predicted the real-world group correctly—similarly, the case of pedestrians 156, 157, and 158 marked red triangles. The pedestrians with ID 159 and 160, represented by a yellow triangle, are a group in the real world, while the model thinks pedestrians with ID 149 are also a part of the group. Similarly, pedestrians with ID 140 and 141 form a group in the real world, but the model does not consider them a group.

6.4. Group split and group merge

It was observed in the video data that groups split, overtake, and merge again in case of oncoming pedestrians, obstacles, and slow-moving pedestrians ahead. One such example is shown in Fig. 7, where a group of pedestrians in the real world split and merge.

Based on this criterion, the model results were analysed again to count the number of group splits and group merges observed (Table 4).

The number of splits and merges increases with the increased value of N. This could be because a higher N value results in more large groups. Thus, the splits and merges reported by the model are higher. Another observation is that the number of splits and merges in clip2 is higher than in clip1. Since the candidate model considers the spatiotemporal parameters, thus the number of splits and merges represents the movement of pedestrians.

7. Conclusion

Pedestrian group detection is a vital task in understanding group dynamics. Earlier studies used spatiotemporal parameters to assess the social bonding between pedestrian pairs, followed by clustering to detect groups. These studies are based on the assumption that social bonding between a pedestrian pair is symmetric, static, and unaffected by surroundings. These assumptions are invalid in the real world. This study proposes an unsupervised group detection method using a candidate-based approach. This study assessed the proposed model qualitatively and evaluated the model quantitatively. For quantitative evaluation, this study proposes three new metrics: GroupRecall, GroupPrecision, and GroupMatchScore, which evaluate the models based on predicted groups instead of pair classification tasks. GroupRecall captures the False negatives, GroupPrecision captures the False positives, and GroupMatchScore captures the exact matches. A visualisation method is developed for qualitative assessment of the group detection model. Group merge and group split are calculated to understand the group behaviour and movement pattern. It was observed that the number of group splits and group merges could explain specific characteristics of pedestrian movement in the crowd.

Since pedestrians are captured for a short period, having a lower time threshold for considering the pedestrian pair for the group is a limitation of this study. This could be overcome by placing the camera higher and capturing pedestrians' movement for a long time. This study considers a long road/corridor geometry, and the proposed model performance can be assessed in different environments, such as open spaces or intersections. Visual cues / psychological parameters can be incorporated into the group detection model to improve the model's performance. Such parameters could also help in understanding other factors influencing the group merge and group split. The study uses real-world trajectories; thus, the model's overall performance would depend on the pedestrian tracking

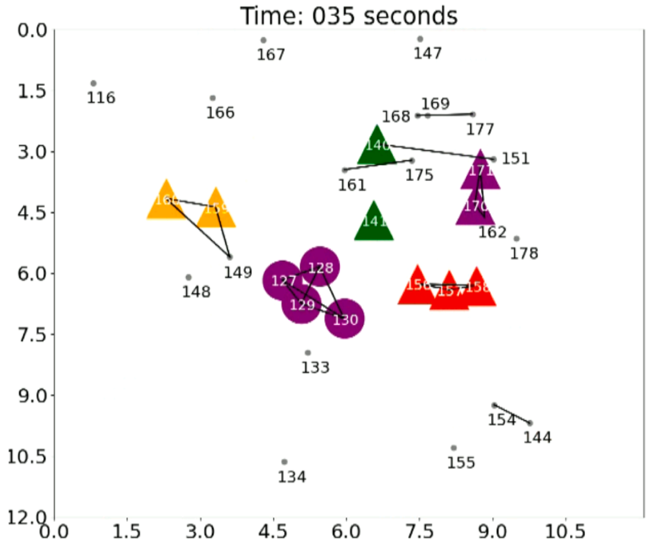
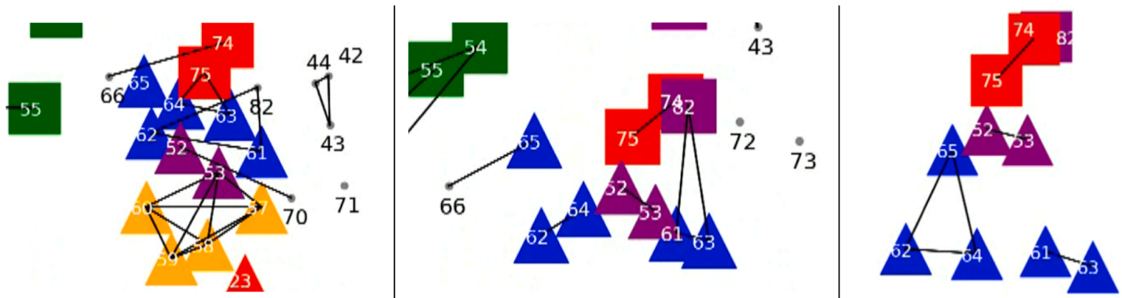


Fig. 6. Snapshot of graph-based visualisation video of candidate model.



(a) Time: 008 seconds (b) Time: 011 seconds (c) Time: 013 seconds

Fig. 7. A snapshot of group split and group merge from real data. *(a) A Group with five pedestrians (ID 61–65) marked with blue triangles. (b) The group got split to overtake pedestrians 52, 53 (c), and then merged again into a single group.*

Table 4
Comparison of the number of splits and merges in clips.

Model	Clip1		Clip2	
	Split	Merge	Split	Merge
Candidate ($N = 3$)	28	41	38	63
Candidate ($N = 4$)	39	53	55	85
Candidate ($N = 5$)	49	65	70	103

model used. A sensitivity test can be carried out to assess model robustness by adding impurity into the trajectories. The study could play a crucial role in exploring other properties of dynamic groups. This would improve the understanding of crowd movement and behaviour, further assisting in crowd management and facility planning.

CRediT authorship contribution statement

Anirban Chakraborty: Writing – review & editing, Supervision, Project administration, Investigation, Conceptualization. **Nipun Choubey:** Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Formal analysis, Conceptualization. **Ashish Verma:** Writing – review & editing, Supervision, Resources, Project administration, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

[1] H. Cheng, Y. Li, M. Sester, Pedestrian group detection in shared Space, in: 2019 *IEEE Intell. Veh. Symp. (IV)*, 2019, pp. 1707–1714, https://doi.org/10.1109/IVS.2019.8813849.
[2] L. Cheng, R. Yarlagadda, C. Fookes, K.D.V.P. Yarlagadda, A review of pedestrian group dynamics and methodologies in modelling pedestrian group behaviours, World J. Mech. Eng. (2014).
[3] N. Choubey, K.P. Shobhana, G. Reddy, A. Verma, Detecting social groups using low mounted camera in mass religious gatherings, Urban Mobil. Res. India (2023).
[4] D.F. Crouse, On implementing 2D rectangular assignment algorithms, IEEE Trans. Aerosp. Electron. Syst. 52 (4) (2016) 1679–1696, https://doi.org/10.1109/TAES.2016.140952.
[5] W. Ge, R.T. Collins, B. Ruback, Autom. Detect. Small Group Struct. a Crowd. (2009). ⟨https://doi.org/10.1109/WACV.2009.5403123⟩.
[6] A. Gorrini, S. Bandini, G. Vizzari, Empirical investigation on pedestrian crowd dynamics and grouping, in: Traffic and Granular Flow '13, Springer International Publishing, 2015, pp. 83–91, https://doi.org/10.1007/978-3-319-10629-8_10.
[7] H.W. Kuhn, The Hungarian method for the assignment problem, in: 50 Years of Integer Programming 1958-2008, Springer Berlin Heidelberg, 2010, pp. 29–47, https://doi.org/10.1007/978-3-540-68279-0_2.
[8] M. Leach, R. Baxter, N. Robertson, E. Sparks, Detecting social groups in crowded surveillance videos using visual attention, in: 2014 IEEE Conf. Comput. Vis. Pattern Recognit. Workshops, 2014, pp. 467–473, https://doi.org/10.1109/CVPRW.2014.75.
[9] M. Li, T. Chen, H. Du, N. Ma, X. Xi, Social group detection is based on multi-level consistent behaviour characteristics, Transp. A: Transp. Sci. (2021) 1–18, https://doi.org/10.1080/23249935.2021.1976877.
[10] L. Lu, C.-Y. Chan, J. Wang, W. Wang, A study of pedestrian group behaviors in crowd evacuation based on an extended floor field cellular automaton model, Transp. Res. Part C: Emerg. Technol. 81 (2017) 317–329, https://doi.org/10.1016/j.trc.2016.08.018.
[11] R. Mazzon, F. Poiesi, A. Cavallaro, Detection and tracking of groups in crowd, in: 2013 *10th IEEE Int. Conf. Adv. Video Signal Based Surveill.*, 2013, pp. 202–207, https://doi.org/10.1109/AVSS.2013.6636640.
[12] R.V.H. Mora-Colque, G. Cámara-Chávez, W. Robson Schwartz, LNCS 8827 - Detect. Groups People Surveill. Videos Based Spatio-Tempo Clues (2014), https://doi.org/10.1007/978-3-319-12568-8_115.
[13] M. Moussaïd, N. Perozo, S. Garnier, D. Helbing, G. Theraulaz, The walking behaviour of pedestrian social groups and its impact on crowd dynamics, PLoS One 5 (4) (2010) e10047, https://doi.org/10.1371/journal.pone.0010047.
[14] C.A.S. Pouw, F. Toschi, F. van Schadewijk, A. Corbetta, Monitoring physical distancing for crowd management: real-time trajectory and group analysis, PLoS One 15 (10) (2020) e0240963, https://doi.org/10.1371/journal.pone.0240963.
[15] M. Schultz, An Individual-based Model for Passenger Movement Behaviour in Airport Terminals, Technische Universität Dresden, 2010.
[16] A. Setia, A. Mittal, Co-operative pedestrians group tracking in crowded scenes using an MST Approach, in: 2015 IEEE Winter Conf. Appl. Comput. Vis., 2015, pp. 102–108, https://doi.org/10.1109/WACV.2015.21.
[17] H. Singh, R. Arter, L. Dodd, P. Langston, E. Lester, J. Drury, Modelling subgroup behaviour in crowd dynamics DEM simulation, Appl. Math. Model. 33 (12) (2009) 4408–4423, https://doi.org/10.1016/j.apm.2009.03.020.
[18] G.H. Subramanian, N. Choubey, A. Verma, Modelling and simulating serpentine group behaviour in crowds using modified social force model, Phys. A: Stat. Mech. Appl. 604 (2022) 127674, https://doi.org/10.1016/j.physa.2022.127674.
[19] G.H. Subramanian, A. Verma, Effect of social groups on crowd movement in mass religious gatherings: a case study of Kumbh Mela, Transp. Dev. Econ. 8 (1) (2022), https://doi.org/10.1007/s40890-021-00145-0.
[20] S. Sun, C. Sun, D.C. Duives, S.P. Hoogendoorn, Neural network model for predicting variation in walking dynamics of pedestrians in social groups, Transportation 50 (3) (2023) 837–868, https://doi.org/10.1007/s11116-021-10263-8.
[21] M.H. Zaki, T. Sayed, Automated analysis of pedestrian group behavior in urban settings, IEEE Trans. Intell. Transp. Syst. 19 (6) (2018) 1880–1889, https://doi.org/10.1109/TITS.2017.2747516.
[22] Y. Zhao, Z. Hu, L. Zhou, M. Liu, J. Liu, Data-driven online group detection based on structured prediction, 2020 IEEE 16th Int. Conf. Autom. Sci. Eng. (CASE) (2020) 1220–1225, https://doi.org/10.1109/CASE48305.2020.9216765.